

Beyond the conjectured bound

Sharp discrepancy for two families of binary substitutions

Counterexamples and corrected theorems for
OEIS A284365 and A284366

with Part II: exact position-error envelopes
for quadratic morphic words (OEIS A284368–A284371)

Research study prepared with ChatGPT

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Abstract

The OEIS entries A284365 and A284366 conjecture that two complementary position sequences associated with the substitution $0 \mapsto 1, 1 \mapsto 101010$ have linear-approximation errors smaller than 2. Both upper bounds are false. Their first counterexamples occur at ranks 8 933 313 and 2 977 771, respectively, and have the same error 2.0048721733261288 We give exact integer certificates for these values and for their minimality. More substantially, we determine the complete discrepancy limit sets for every substitution $0 \mapsto 1, 1 \mapsto (10)^m$, $m \geq 1$. For the OEIS case $m = 3$, the sharp common upper bound is $(6 + \sqrt{21})/5 = 2.116515138991168 \dots$. The proof uses two state-dependent prefix-weight intervals and a contracting set recursion. We also prove the companion density conjectures, construct explicit infinite families of counterexamples, derive rational generating functions for their indices, and provide exact-arithmetic random-access and verification software. The mathematical proofs are self-contained; finite experiments are supporting checks, not substitutes for the proofs.

Part II, added on October 1, 2026 (batch 73O1), applies the same method to the disjoint family $0 \mapsto 1, 1 \mapsto 10^a 1^b$ with $a, b \geq 1$. It proves that prefix discrepancy is bounded exactly when $a \leq b + 1$ (logarithmic growth at $a = b + 2$, power growth beyond), determines the four exact state extrema and the sharp, unattained position-error envelopes throughout the bounded regime, and specializes them to the OEIS words A284368 ($1 \mapsto 1011$) and A284369 ($1 \mapsto 1001$). For A284369 this proves the conjectured bounds of A284370 and A284371 with sharper constants; for A284368 it recovers, as a calibration, the known s -Wythoff floor formulas of A184484–A184485.

Status and scope. The cited OEIS pages still labeled the relevant statements as conjectures when accessed on September 19, 2026. This article establishes the stated results by explicit proofs and executed certificates. It does not claim an exhaustive priority search, peer review, or proof-assistant formalization. General spectral methods for substitution discrepancy are classical; the contribution here is the exact solution and certificates for this family and these particular conjectures.

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Part I. Sharp discrepancy for a family of binary substitutions

1 The selected problem and its resolution

Let σ be the substitution

$$\sigma(0) = 1, \quad \sigma(1) = 101010. \quad (1.1)$$

Its fixed word beginning with 1 is

$$\mathbf{w} = 101010110101011010101 \dots$$

This is A284364 [1]. Number positions starting at 1. Let $u(n)$ and $v(n)$ be the positions of the n th zero and the n th one, respectively. Thus u is A284365 and v is A284366 [2, 3]. Put

$$r = \frac{9 + \sqrt{21}}{6}, \quad s = \frac{\sqrt{21} - 1}{2}, \quad \frac{1}{r} + \frac{1}{s} = 1. \quad (1.2)$$

The position-sequence entries conjecture

$$-2 < rn - u(n) < 2, \quad -1 < sn - v(n) < 2 \quad (n \geq 1). \quad (1.3)$$

They also conjecture their respective densities to be $(9 - \sqrt{21})/10$ and $(1 + \sqrt{21})/10$. The density comments are dated April 13, 2025; the sequence entries credit Clark Kimberling, March 26, 2017. We use the unambiguous position-sequence formulations: the parent entry's comment uses count terminology alongside position slopes, which would be a different and immediately incompatible statement.

Theorem 1.1 (Resolution for the OEIS sequences). Write $E_0(n) = rn - u(n)$ and $E_1(n) = sn - v(n)$. For every $n \geq 1$,

$$-\frac{12 + 2\sqrt{21}}{15} < E_0(n) < \frac{6 + \sqrt{21}}{5}, \quad (1.4)$$

$$\frac{\sqrt{21} - 9}{5} < E_1(n) < \frac{6 + \sqrt{21}}{5}. \quad (1.5)$$

Each endpoint is a subsequential limit, and indeed the full set of subsequential limits of each error sequence is the corresponding closed interval. In particular, both conjectured lower bounds in (1.3) hold, but both upper bounds fail infinitely often.

The first counterexamples are

$$v(2\,977\,771) = 5\,334\,043, \quad u(8\,933\,313) = 20\,222\,898. \quad (1.6)$$

Their common error is

$$\frac{2\,977\,771\sqrt{21} - 13\,645\,857}{2} = 2.00487217332612884794005732278\dots > 2. \quad (1.7)$$

The two asserted densities are correct.

The bounds are not consequences of extrapolating the observed maxima. Sections 2–5 prove them for an entire family. Section 6 supplies a finite certificate for (1.6) and its minimality. Section 7 gives an explicit sequence of errors increasing to the sharp upper endpoint. The method belongs to the established study of the relation between substitution spectra and discrepancy [4]; all specific interval and sharpness arguments needed here are proved below.

2 A parameterized family and its discrepancy theorem

Fix an integer $m \geq 1$ and define

$$\sigma_m(0) = 1, \quad \sigma_m(1) = (10)^m. \quad (2.1)$$

The image of 1 begins with 1 and has length at least 2, so the words $\sigma_m^k(1)$ form nested prefixes of an infinite fixed word $\mathbf{w}^{(m)}$. The substitution contains both letters in an iterate of either letter; no additional initial condition is needed beyond the initial 1.

Throughout the general argument let

$$q = \frac{\sqrt{m^2 + 4m} - m}{2}, \quad \lambda = m + q = \frac{m}{q}, \quad s = 1 + q, \quad r = 1 + \frac{1}{q}, \quad (2.2)$$

and

$$t = 1 - q, \quad C = \frac{q}{1 - q^2}, \quad H = qC = \frac{q^2}{1 - q^2}. \quad (2.3)$$

The quantities depend on m , even when the subscript is suppressed. The following identities will be used repeatedly:

$$0 < q < 1, \quad q^2 = m(1 - q), \quad q(1 + H) = C, \quad H(1 + q) = m, \quad \frac{1}{r} + \frac{1}{s} = 1. \quad (2.4)$$

For example, $H(1 + q) = q^2/(1 - q) = m$. Also q is irrational: $(m + 1)^2 < m^2 + 4m < (m + 2)^2$ for $m \geq 1$.

Let $Z(N)$ and $O(N)$ count the zeros and ones in the first N letters, with $Z(0) = O(0) = 0$. Let $u_m(n)$ and $v_m(n)$ denote the corresponding position sequences, and set

$$E_0(n) = rn - u_m(n), \quad E_1(n) = sn - v_m(n).$$

Theorem 2.1 (Sharp error intervals for every m). *For every integer $m \geq 1$ and $n \geq 1$,*

$$C - m < E_1(n) < C, \quad (2.5)$$

$$\frac{1 - C}{q} < E_0(n) < C. \quad (2.6)$$

The closures of the two error-value sets are exactly these intervals with their endpoints included. These are also their full subsequential limit sets. No endpoint is attained at a finite index.

For every $N \geq 0$,

$$-\frac{H}{1 + q} < O(N) - \frac{N}{1 + q} < \frac{C}{1 + q}. \quad (2.7)$$

Again, the two endpoints are sharp, and the closure of the centered counting errors is the full closed interval. In particular,

$$\lim_{N \rightarrow \infty} \frac{O(N)}{N} = \frac{1}{s}, \quad \lim_{N \rightarrow \infty} \frac{Z(N)}{N} = \frac{1}{r}.$$

A useful equivalent form of the lower zero-position endpoint is

$$\frac{1 - C}{q} = C - 1 - \frac{m - 1}{q}. \quad (2.8)$$

For $m = 3$, substituting $q = (\sqrt{21} - 3)/2$ gives

$$C = \frac{6 + \sqrt{21}}{5}, \quad H = \frac{3 + 3\sqrt{21}}{10},$$

and (2.5)–(2.6) become (1.5)–(1.4).

3 A contracting weight and two prefix states

For a finite binary word P , define its weight by

$$X(P) = |P|_0 - q|P|_1. \quad (3.1)$$

This is additive under concatenation. The incidence matrix of the substitution, with row order 0, 1 and column order 0, 1, is

$$M = \begin{pmatrix} 0 & m \\ 1 & m \end{pmatrix}.$$

Its eigenvalues are λ and $-q$. More directly, the letter weights give the identity on which the proof rests:

Lemma 3.1 (Weight contraction). *For every finite word P ,*

$$X(\sigma_m(P)) = -qX(P). \quad (3.2)$$

Proof. For $P = 0$ the two sides equal $-q$. For $P = 1$, the image has m letters of each type and hence weight $m(1 - q) = q^2 = -q(-q)$. Additivity proves the claim for an arbitrary concatenation of letters. $\square$

The next letter is crucial. For $a \in \{0, 1\}$, let

$$S_a = \{X(w_1 \cdots w_N) : N \geq 0, w_{N+1} = a\}. \quad (3.3)$$

Thus $0 \in S_1$, from the empty prefix. A prefix ending immediately before a zero belongs to state 0, not to a state determined by its last letter.

Lemma 3.2 (Exact state equations). *The sets in (3.3) satisfy*

$$S_0 = \bigcup_{j=0}^{m-1} (-qS_1 + jt - q), \quad (3.4)$$

$$S_1 = (-qS_0) \cup \bigcup_{j=0}^{m-1} (-qS_1 + jt). \quad (3.5)$$

Proof. View the fixed word as the concatenation of images of its own letters. If a parent prefix P is followed by 0, its image is followed by the single letter 1, giving the transition $-qS_0$ in (3.5). If P is followed by 1, its image is followed by $(10)^m$. Immediately before the $(j + 1)$ st one in that image, the internal prefix is $(10)^j$, whose weight is jt . Immediately before the corresponding zero, the internal prefix is $(10)^j1$, whose weight is $jt - q$. Lemma 3.1 gives all the stated affine images. Conversely, every location in the fixed word lies in one of these image blocks. Thus there are no missing transitions and the equations are set equalities. $\square$

Using a single state and adding all possible offsets independently would forget which parent letter permits which offset. The two-state system retains precisely the information required for the sharp constants.

4 Solving the interval recursion, including sharpness

Define two closed intervals

$$I_0 = [-C, H - 1], \quad I_1 = [-qH, H]. \quad (4.1)$$

They are nondegenerate for every $m \geq 1$.

Lemma 4.1 (Invariant intervals). *The pair (I_0, I_1) satisfies (3.4)–(3.5) with S_a replaced by I_a .*

Proof. Put $J_j = -qI_1 + jt$. Each J_j has length

$$qH(1 + q) = mq,$$

while successive left endpoints are separated by $t = 1 - q = q^2/m$. If $m \geq 2$, then $mq \geq t$, so adjacent intervals overlap. For $m = 1$ there is only one interval. The extreme endpoints yield

$$\bigcup_{j=0}^{m-1} J_j = [-qH, H - t]. \quad (4.2)$$

Indeed, the upper endpoint is $q^2H + (m - 1)t = H - t$, since $q^2(H + 1) = H$ and $mt = q^2$.

Subtracting q from (4.2) gives

$$\bigcup_{j=0}^{m-1} (J_j - q) = [-qH - q, H - t - q] = [-C, H - 1] = I_0.$$

For state 1, the other image is

$$-qI_0 = [q - qH, H].$$

It overlaps or touches the interval in (4.2), because

$$H - t \geq q - qH \iff H(1 + q) \geq 1 \iff m \geq 1.$$

Their union is therefore $[-qH, H] = I_1$. □

Lemma 4.2 (Strict inclusion of all actual prefixes). *Every element of S_a lies in the interior of I_a .*

Proof. We induct on the length N of the prefix. The empty prefix has weight 0, which lies strictly inside I_1 .

For a nonempty prefix, locate its next letter in an image block. The associated parent prefix has some length $k < N$. To see strict decrease, if $k > 0$ then that parent prefix begins with 1, so its image has length at least $k + (2m - 1) > k$; this image is contained in the original prefix. If $k = 0$, strict decrease follows from $N > 0$.

By induction, the parent weight belongs to the interior of its appropriate interval. Its actual transition is an affine map with nonzero slope $-q$. The image of an interior point is an interior point of the image interval, and Lemma 4.1 puts that image interval inside the appropriate target interval. Such a point cannot be an endpoint of the target interval. This completes the induction. □

Bounds alone do not prove sharpness. We next show that the intervals contain no gap in the closures of the actual prefix weights.

Lemma 4.3 (Uniqueness for the compact set recursion). *Let K_0, K_1 be nonempty compact subsets of $\mathbb{R}$ satisfying the state equations. Then $K_0 = I_0$ and $K_1 = I_1$.*

Proof. For compact nonempty subsets of $\mathbb{R}$, write d_H for Hausdorff distance. An affine map of slope $-q$ multiplies this distance by q . Also, for finite indexed families,

$$d_H\left(\bigcup_i A_i, \bigcup_i B_i\right) \leq \max_i d_H(A_i, B_i).$$

Let $D = \max\{d_H(K_0, I_0), d_H(K_1, I_1)\}$. Apply these observations to the two state equations for K and I . They imply $D \leq qD$. Since D is finite and $0 < q < 1$, it follows that $D = 0$. $\square$

Proposition 4.4 (Exact closures).

$$\text{cl } S_0 = [-C, H - 1], \quad \text{cl } S_1 = [-qH, H]. \quad (4.3)$$

The closure of all prefix weights is $[-C, H]$.

Proof. Lemma 4.2 makes S_0, S_1 bounded, and both are nonempty. Take closures in their exact finite-union equations. Closure commutes with each affine map and with a finite union, so $(\text{cl } S_0, \text{cl } S_1)$ is a compact fixed pair. Lemma 4.3 proves (4.3). Finally I_0 and I_1 overlap or touch, since $H - 1 \geq -qH$ is again equivalent to $m \geq 1$. Their union is $[-C, H]$. $\square$

Distinct prefix lengths have distinct weights. Indeed, equality would give $\Delta Z = q\Delta O$, impossible for irrational q unless both integer differences vanish, in which case the lengths agree. A set dense in a nondegenerate interval has infinitely many distinct points in every relative neighborhood of every point of that interval. Consequently its density statements remain valid after deleting any finite number of prefixes. This proves that the closures just obtained are also the full sets of limits along prefix lengths tending to infinity.

5 Position errors, densities, and exact transfer

5.1 Passing from prefix weights to position errors

Immediately before the n th one, the prefix has $n - 1$ ones and $v_m(n) - n$ zeros. Its weight is therefore

$$X = v_m(n) - n - q(n - 1) = q - E_1(n). \quad (5.1)$$

Immediately before the n th zero, the prefix has $n - 1$ zeros and $u_m(n) - n$ ones, giving

$$X = n - 1 - q(u_m(n) - n) = qE_0(n) - 1. \quad (5.2)$$

Apply the strict state bounds and Proposition 4.4. Equation (5.1) transforms I_1 into

$$[q - H, q + qH] = [C - m, C],$$

and (5.2) transforms I_0 into $[(1 - C)/q, C]$. The identities follow from (2.4). The state sets contain exactly the prefixes immediately before the respective ranked letters, so no additional selection restriction is needed. This proves the position part of Theorem 2.1, including strictness and the full subsequential limit sets.

For ordinary prefix counts,

$$X = N - (1 + q)O(N), \quad O(N) - \frac{N}{1 + q} = -\frac{X}{1 + q}. \quad (5.3)$$

The total prefix-weight interval $[-C, H]$ proves (2.7) and its sharpness. Dividing the bounded counting error by N establishes the densities without invoking an ergodic theorem. In the case $m = 3$, this yields the quantitative estimate

$$-\frac{33 + 3\sqrt{21}}{50} < O(N) - \frac{1 + \sqrt{21}}{10}N < \frac{27 + 7\sqrt{21}}{50}. \quad (5.4)$$

Thus the density conjectures in all three relevant OEIS entries follow, and the error in the unnormalized counting function is uniformly bounded.

5.2 A relation between the two position sequences

Proposition 5.1 (Exact position transfer). *For $n \geq 1$ and $1 \leq h \leq m$,*

$$u_m(m(n-1) + h) = v_m(n) + (2m-1)n - 2m + 2h. \quad (5.5)$$

Equivalently,

$$E_0(m(n-1) + h) = E_1(n) - (m-h)(r-2). \quad (5.6)$$

In particular,

$$u_m(mn) = v_m(n) + (2m-1)n, \quad E_0(mn) = E_1(n). \quad (5.7)$$

Proof. Before the parent word's n th one there are $n-1$ ones and $v_m(n) - n$ zeros. The total length of their images is $2m(n-1) + v_m(n) - n$. The image of the next one contains m zeros, at internal positions $2, 4, \dots, 2m$, and no zero comes from the image of a parent zero. Thus the h th such zero has global rank $m(n-1) + h$ and the position in (5.5).

Subtract this expression from $r(m(n-1) + h)$. The identity $m(r-2) = q$ reduces the result to (5.6). $\square$

For $m = 3$, $C < r$; explicitly

$$r - C = \frac{9 - \sqrt{21}}{30} > 0.$$

Since $r - 2 > 0$, (5.6) shows that the errors at zero ranks $3(n-1) + 1$ and $3(n-1) + 2$ are strictly less than $C - (r-2) < 2$. It follows that

$$\{k : E_0(k) > 2\} = \{3n : E_1(n) > 2\}. \quad (5.8)$$

Equality to 2 is impossible, because r and s are irrational. This bijection will make the zero-position minimality result immediate once the first one-position violation has been certified.

6 An exact finite certificate and the first counterexamples

6.1 Recursive blocks

Define auxiliary finite words

$$W_{-1} = 0, \quad W_0 = 1, \quad W_k = (W_{k-1}W_{k-2})^m \quad (k \geq 1). \quad (6.1)$$

For $k \geq 0$, induction gives $W_k = \sigma_m^k(1)$. In particular, W_k is a prefix of $\mathbf{w}^{(m)}$. Write L_k, Z_k, O_k for its length, zero count, and one count. Each satisfies

$$a_k = m(a_{k-1} + a_{k-2}). \quad (6.2)$$

For the remainder of this section $m = 3$.

Consider the finite word

$$P = W_{11} W_{10} W_{10} W_8 W_6 W_4 W_2. \quad (6.3)$$

This is a prefix of W_{12} , followed by a 1. Here is an explicit way to follow the recursive decomposition. In $W_{12} = (W_{11}W_{10})^3$, take the first $W_{11}W_{10}$, then enter the next $W_{11} = (W_{10}W_9)^3$ and take its first W_{10} . The residual word begins with W_9 . In $W_9 = (W_8W_7)^3$, take W_8 , leaving a word beginning with W_7 . Repeat with W_7, W_5, W_3 , taking W_6, W_4, W_2 . The remaining word begins with $W_1 = 101010$, so its next letter is 1.

The count recurrence gives

$$|P|_0 = 2\,356\,272, \quad |P|_1 = 2\,977\,770, \quad |P| = 5\,334\,042. \quad (6.4)$$

Consequently the next one proves the first identity in (1.6). Formula (5.7) proves the second:

$$u(3 \cdot 2\,977\,771) = 5\,334\,043 + 5 \cdot 2\,977\,771 = 20\,222\,898.$$

Their common error is exactly (1.7). To certify that it exceeds 2, it suffices to observe

$$21(2\,977\,771)^2 - (13\,645\,861)^2 = 265\,940 > 0. \quad (6.5)$$

Both quantities compared before squaring are positive, so (6.5) proves $2\,977\,771\sqrt{21} > 13\,645\,861$ with no numerical approximation.

6.2 Why no earlier index is a counterexample

For any finite word U , define

$$\mu_1(U) = \min\{X(Q) : Q \text{ is a proper prefix of } U, \text{ and the next letter in } U \text{ is } 1\}.$$

Use $+\infty$ if U has no one. For concatenations,

$$\mu_1(UV) = \min\{\mu_1(U), X(U) + \mu_1(V)\}, \quad (6.6)$$

$$\mu_1(B^m) = \mu_1(B) + \min\{0, (m-1)X(B)\}. \quad (6.7)$$

The second formula follows by considering the m translated copies of B . These identities give a small exact certificate for all earlier positions at once.

Apply (6.6) to the seven factors of P . The translated candidate minima, in their order of appearance, are

$$\begin{aligned} &431\,712 - 545\,583q, & 1\,820\,700 - 2\,300\,934q, \\ &2\,222\,379 - 2\,808\,561q, & 2\,346\,957 - 2\,965\,998q, \\ &2\,355\,624 - 2\,976\,951q, & 2\,356\,227 - 2\,977\,713q, \\ &2\,356\,269 - 2\,977\,766q. \end{aligned} \quad (6.8)$$

Their minimum is the first, so

$$\mu_1(P) = 431\,712 - 545\,583q. \quad (6.9)$$

All of these finite comparisons can be checked by the integer sign rule in Section 8; the separate checker in the archive recomputes the block table and its minimum rather than accepting them as input.

k	L_k	Z_k	O_k	$\mu_1(W_k)$
0	1	0	1	0
1	6	3	3	0
2	21	9	12	$6 - 8q$
3	81	36	45	$9 - 12q$
4	306	135	171	$99 - 126q$
5	1 161	513	648	$144 - 183q$
6	4 401	1 944	2 457	$1 440 - 1 821q$
7	16 686	7 371	9 315	$2 088 - 2 640q$
8	63 261	27 945	35 316	$20 718 - 26 184q$
9	239 841	105 948	133 893	$30 033 - 37 956q$
10	909 306	401 679	507 627	$297 819 - 376 374q$
11	3 447 441	1 522 881	1 924 560	$431 712 - 545 583q$
12	13 070 241	5 773 680	7 296 561	$4 280 832 - 5 409 957q$

Table 1: Block data for $m = 3$. Every minimum is calculated by (6.6)–(6.7), using exact quadratic-field comparisons. The $k = 12$ row is included for context; only the preceding rows are needed to certify P .

By (5.1), the largest one-position error strictly before the candidate equals

$$q - \mu_1(P) = 545\,584q - 431\,712 = 1.988978393507077166 \dots < 2. \quad (6.10)$$

The last inequality has the exact certificate

$$(2\,500\,180)^2 - 21(545\,584)^2 = 110\,224 > 0. \quad (6.11)$$

It proves $545\,584\sqrt{21} < 2\,500\,180$, which is precisely (6.10) < 2 after substituting the value of q . The maximum in (6.10) occurs at rank 545 584, at position 977 296.

Every earlier ranked one lies in P . Hence no earlier one-position error is at least 2. Equations (6.5) and (6.11) certify the first counterexample, and the exact bijection (5.8) certifies the first zero-position counterexample. This proof covers all earlier indices; it is not a claim based only on a finite sample of selected positions.

7 Infinite extremal families and their generating functions

The compact-set proof gives sharpness nonconstructively in terms of closures. A specific family approaching the common upper endpoint can also be written down and evaluated by short recurrences.

Proposition 7.1 (Explicit extremal prefixes). *For general $m \geq 1$, let P_0 be the empty word and define*

$$P_{j+1} = \sigma_m^2(P_j)(10)^m, \quad U_j = \sigma_m(P_j). \quad (7.1)$$

Every P_j and every U_j is a prefix of the fixed word, followed by a 1. Set

$$n_j = |U_j|_1 + 1, \quad p_j = |U_j| + 1. \quad (7.2)$$

Then

$$v_m(n_j) = p_j, \quad E_1(n_j) = C(1 - q^{2j+2}) \nearrow C. \quad (7.3)$$

The zero-position errors at ranks mn_j are equal to these errors.

Proof. Suppose P_j is a prefix followed by 1. Its image is followed by $(10)^m$. Therefore $\sigma_m(P_j)1$ is a prefix followed by 0. Apply the substitution again: the resulting prefix is $\sigma_m^2(P_j)(10)^m = P_{j+1}$, followed by $\sigma_m(0) = 1$. This proves the prefix assertion inductively from P_0 . It also shows that U_j is followed by 1.

By Lemma 3.1 and $m(1 - q) = q^2$,

$$X(P_{j+1}) = q^2 X(P_j) + q^2.$$

Starting at zero and summing the geometric progression gives $X(P_j) = H(1 - q^{2j})$. Thus

$$E_1(n_j) = q - X(U_j) = q + qH(1 - q^{2j}) = C - Cq^{2j+2},$$

where $q + qH = C$ and $qH = Cq^2$. The errors increase strictly because $0 < q < 1$. The zero-position assertion follows from (5.7). $\square$

Let $(z_j, o_j) = (|P_j|_0, |P_j|_1)$. The preceding construction is fully integer-recursive:

$$\begin{pmatrix} z_{j+1} \\ o_{j+1} \end{pmatrix} = \begin{pmatrix} m & m^2 \\ m & m^2 + m \end{pmatrix} \begin{pmatrix} z_j \\ o_j \end{pmatrix} + \begin{pmatrix} m \\ m \end{pmatrix}, \quad (z_0, o_0) = (0, 0), \quad (7.4)$$

with

$$n_j = z_j + mo_j + 1, \quad p_j = z_j + 2mo_j + 1. \quad (7.5)$$

Eliminating the two counts yields an especially simple result.

Corollary 7.2 (Scalar recurrences and rational generating functions). *Put $T = m(m + 2)$. Then*

$$n_{j+2} = Tn_{j+1} - m^2n_j + 1 - m, \quad n_0 = 1, \quad n_1 = m^2 + m + 1, \quad (7.6)$$

$$p_{j+2} = Tp_{j+1} - m^2p_j + m^2 - m + 1, \quad p_0 = 1, \quad p_1 = 2m^2 + m + 1. \quad (7.7)$$

Their ordinary generating functions are

$$\sum_{j \geq 0} n_j z^j = \frac{1 - mz}{(1 - z)(1 - m(m + 2)z + m^2 z^2)}, \quad (7.8)$$

$$\sum_{j \geq 0} p_j z^j = \frac{1 + m(m - 1)z}{(1 - z)(1 - m(m + 2)z + m^2 z^2)}. \quad (7.9)$$

Proof. The linear part in (7.4) is M^2 , with trace T and determinant m^2 . Its characteristic identity implies the two inhomogeneous recurrences after substitution in (7.5); alternatively, one may eliminate z_j, o_j directly. The constant terms and initial values follow by evaluating $j = 0, 1, 2$.

For a sequence obeying $a_{j+2} = Ta_{j+1} - m^2a_j + c$, multiplying its series by $1 - Tz + m^2z^2$ gives $a_0 + (a_1 - Ta_0)z + cz^2/(1 - z)$. For (7.6), the numerator after multiplication by $1 - z$ reduces to $1 - mz$; for (7.7), it reduces to $1 + m(m - 1)z$. This proves both generating functions. $\square$

In the OEIS case the recurrences are $n_{j+2} = 15n_{j+1} - 9n_j - 2$ and $p_{j+2} = 15p_{j+1} - 9p_j + 7$. Formula (7.3) shows that every $j \geq 6$ gives a counterexample. The closed-interval theorem is stronger still: every open interval contained in $(2, C)$ contains infinitely many errors of each position sequence, not just those in this special family.

j	n_j	$p_j = v(n_j)$	$E_1(n_j)$, rounded
0	1	1	0.791287847478
1	13	22	1.286742017213
2	184	328	1.596963935937
3	2 641	4 729	1.791205189187
4	37 957	67 990	1.912826719410
5	545 584	977 296	1.988978393507
6	7 842 145	14 047 537	2.036659732964
7	112 721 917	201 917 398	2.066514757944
8	1 620 249 448	2 902 333 144	2.085208077525

Table 2: An explicit extremal family for $m = 3$. It supplies infinitely many counterexamples, but its first violation ($j = 6$) is not the first violation over all ranks.

7.1 Exact asymptotics along the extremal family

The denominator in (7.8)–(7.9) factors as

$$(1 - z)(1 - \lambda^2 z)(1 - q^2 z).$$

Here $\lambda > 1$ and $0 < q < 1$, so partial fractions give an exact three-term expansion. For either sequence $a_j = n_j$ or $a_j = p_j$, let

$$d = \frac{c}{1 - 2m}, \quad A = \frac{a_1 - d - q^2(a_0 - d)}{\lambda^2 - q^2}, \quad B = a_0 - d - A,$$

where c is the corresponding constant in its recurrence. Then

$$a_j = A\lambda^{2j} + d + Bq^{2j}. \quad (7.10)$$

The leading coefficient is positive, as follows either from these expressions or the positive count recurrence. In particular, $n_j, p_j = \Theta(\lambda^{2j})$, whereas

$$C - E_1(n_j) = Cq^{2j+2} = \Theta(p_j^{-\kappa}), \quad \kappa = -\frac{\log q}{\log \lambda}. \quad (7.11)$$

For $m = 3$, $\kappa = 0.17565278149 \dots$. This is a statement along the constructed subsequence, not a claim about the optimal record-error convergence rate over every prefix. The comparatively small exponent explains why substantial growth in the rank can coexist with a slow approach to the limiting extremum.

8 Exact algorithms and executed checks

8.1 Quadratic arithmetic without rounding decisions

Every prefix weight and every one-position error is of the form $a + bq$ with integer a, b . Since $q^2 = m - mq$, addition and multiplication stay in this representation. To compare $a + bq$ with zero, put

$$A = 2a - mb, \quad D = m^2 + 4m.$$

Its sign is the sign of $A + b\sqrt{D}$. If A and b have the same nonzero sign, that sign is the answer. Zero cases are immediate. If they have opposite signs, the answer is

$$\text{sgn}(A) \text{sgn}(A^2 - Db^2). \quad (8.1)$$

As D is nonsquare, there is no unhandled nontrivial equality case. Thus all block-extremum decisions and counterexample inequalities in the software reduce to integer arithmetic. Zero-position errors may contain $1/q$; multiplying their comparisons by the positive number q puts them in the same representation.

8.2 Associative summaries and random access

A block summary stores its zero and one counts, and for each following letter state the minimum and maximum prefix weight, together with a location attaining the extremum. Under concatenation UV , the extrema from V are translated by $X(U)$ and their positions by $|U|$; the appropriate minimum or maximum is then taken with the extrema from U . This is the same identity as (6.6) in both states and both directions.

Precompute summaries for $W_{-1}, W_0, \dots, W_k$ using (6.1). To count the first N letters, find the first block long enough, skip whole child pairs by their lengths, and descend into the residual child. To find the n th specified letter, use its block counts instead of lengths. At a fixed m , there are $O(\log N)$ levels, and each level involves only a bounded number of arithmetic operations. The library groups repeated child pairs for these operations.

To find the first error at least a threshold B , a block beginning after a prefix of weight x can be discarded exactly when its appropriate extremum cannot reach the threshold. For ones the feasibility condition is

$$x + \mu_1(W_k) \leq q - B.$$

For zeros, using the maximum prefix weight $\nu_0(W_k)$ before a zero, the condition is

$$x + \nu_0(W_k) \geq qB - 1.$$

Find a first feasible enclosing block, then inspect children from left to right and descend into the first feasible child, skipping the others by their counts. The returned leaf is the first violation, and the skipped block extrema certify that no earlier one was missed.

For fixed m and a counterexample at a position of size N , the precomputation and descent use $O(\log N)$ arithmetic operations on $O(\log N)$ -bit integers. This is an arithmetic-operation count, not a unit-cost bit-complexity assertion. With $M(b)$ denoting the cost of multiplying b -bit integers, the sign tests give the conservative bound $O(\log N M(\log N))$ bit operations, alongside additions and small-parameter divisions. Storing all summaries uses $O((\log N)^2)$ bits at fixed m . Searching only up to a prescribed level and finding nothing is a finite negative result, not a proof of nonexistence in the infinite word.

8.3 What was actually checked

The delivered script code/verify.py -long completed successfully under Python 3.13.5. It recorded 310 816 explicit checks, in addition to the assertions internal to the separate certificate checker. For every $m = 1, \dots, 8$, it built a literal 12 000-letter prefix, checked the sharp bounds at every letter, and compared selected random-access queries and complete small-block extrema against direct calculations. It also checked the transfer identity, the exact extremal family through $j = 24$, and the proposed recurrences and generating-function numerators.

For $m = 3$, the script compared the first 30 displayed terms of each position sequence and the first 30 binary letters with the OEIS entries. It did not download or compare the complete linked 10 000-term tables. The optional long test independently formed the fixed-word prefix of length 20 222 898 by literal substitution, confirming both specified counterexample positions and ranks. The first-counterexample search used exact summaries, not a floating-point scan.

The file code/`independent_check.py` does not import the main library. It reconstructs the count and minimum certificate with a separate integer sign test and concatenation implementation. This checks the finite computations supporting minimality; the prefix-membership argument and transfer theorem supply their mathematical justification. The verification data in `data/` records exact coefficients as well as display decimals, and includes the full first-violation descent traces. No experiment is used to infer the infinite interval theorem.

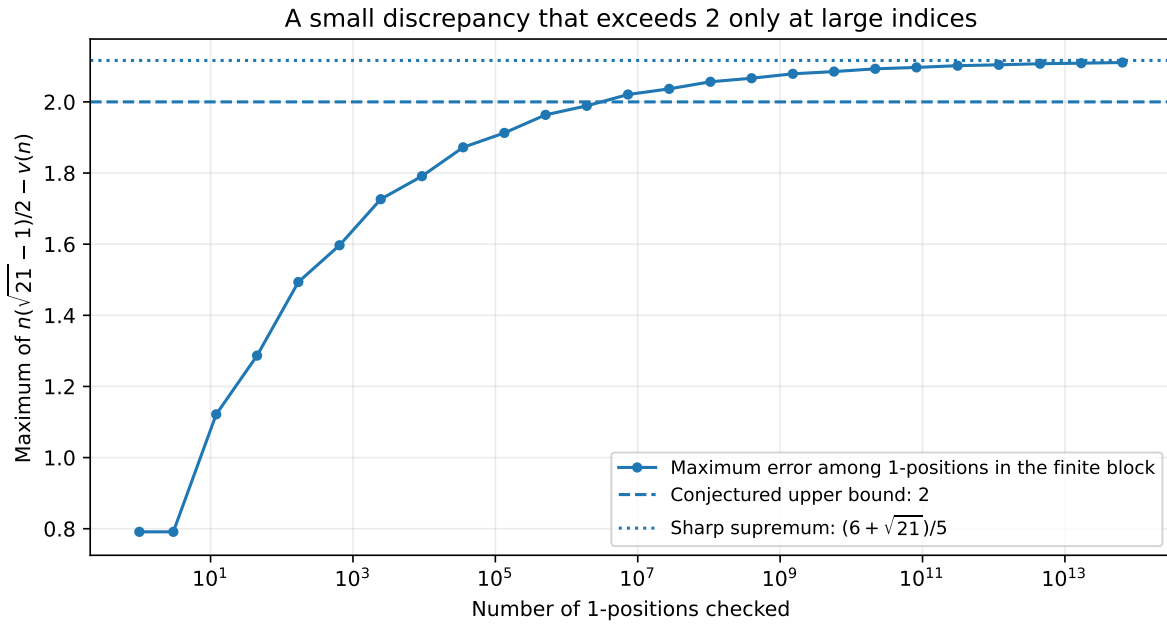


Figure 1: Maximum one-position error over all ones in W_k , for $k = 0, \dots, 24$. The horizontal scale is the number of one-positions in the enclosing block, not the rank at which its maximum is attained. The curve is computed from exact block summaries; floating-point values are used only to render the figure.

9 Further consequences and boundaries of the result

9.1 The $m = 1$ endpoint is a Beatty pair

At $m = 1$, $q = (\sqrt{5} - 1)/2$ and $C = 1$. Both error intervals become $(0, 1)$. Since the positions are integers and the slopes are irrational,

$$v_1(n) = \lfloor (1 + q)n \rfloor, \quad u_1(n) = \left\lfloor \left(1 + \frac{1}{q}\right)n \right\rfloor.$$

Thus the familiar complementary Beatty description is recovered directly from the theorem. For $m \geq 2$, no fixed real shift γ can give $v_m(n) = \lfloor sn + \gamma \rfloor$ for every n : such an expression would put all its errors in an interval of length 1, whereas the error closure in (2.5) has length exactly m . The density already forces any putative Beatty slope to be s .

9.2 A finite factor-balance bound

The centered prefix-counting errors have range width

$$\frac{C + H}{1 + q} = C.$$

For a factor of length L , its number of ones minus L/s is the difference of two prefix errors, so it lies strictly between $-C$ and C . Two factors of the same length consequently differ in their numbers of ones by an integer whose absolute value is strictly less than $2C$. Hence

$$||A|_1 - |B|_1| \leq \lceil 2C \rceil - 1 \quad \text{whenever } |A| = |B|. \quad (9.1)$$

For $m = 3$ this gives a balance bound of 4. The article does not assert that this bound is optimal: sharp prefix extrema do not automatically occur at two endpoints separated by the same prescribed factor length. The exact optimal factor-balance constant is a separate question.

9.3 The word itself has a nonrational generating function

Although the extremal-index series (7.8)–(7.9) are rational, the characteristic series

$$F_m(z) = \sum_{n \geq 1} w_n^{(m)} z^n$$

is not rational. Indeed, a rational power series has coefficients obeying an eventual fixed linear recurrence. If those coefficients take values in a finite set, the vector of a fixed number of consecutive coefficients has only finitely many states and deterministically produces its successor. A state eventually repeats, making the sequence eventually periodic. An eventually periodic binary sequence has rational density, contradicting $1/(1+q)$, which is irrational. This elementary observation concerns the full word, not the rational generating functions of the sparse families.

9.4 What remains outside the claims

The conjectured upper bounds and density statements selected in Section 1 are resolved by the preceding proofs. The article does not determine the frequency distribution of discrepancy values within their interval, the density of the ranks with error above 2, or the exact best factor-balance constant. Knowing the support of subsequential limits does not by itself answer those distributional questions. Nor is the constructed extremal subsequence claimed to enumerate all record errors. These distinctions prevent stronger statements from being inferred merely from the observed finite data or the interval closure result. [Added 1 October 2026, batch 73O1: Part II (research question 3 of Section 21) asks the same density question for the family $0 \mapsto 1$, $1 \mapsto 10^a 1^b$; it remains open in both Parts. The scope of Part I is the family $1 \mapsto (10)^m$ only; Part II treats the disjoint family $1 \mapsto 10^a 1^b$ by the same method (Remark 11.1).]

10 Conclusion

A conjecture about an error below 2 can survive a substantial finite check while failing only slightly at a much larger index. Here the first one-position violation occurs at rank 2 977 771, and its transfer produces the first zero-position violation at rank 8 933 313. The finite disproof is certified by a short recursive prefix and two integer square comparisons. The structural analysis goes further: the correct common upper bound is $(6 + \sqrt{21})/5$, and a two-state contraction proves exact discrepancy intervals for every $0 \mapsto 1$, $1 \mapsto (10)^m$. Density, explicit extremal families, rational index-generating functions, and efficient exact verification then follow from the same framework.

Part II. Exact position-error envelopes for quadratic morphic words

A two-parameter discrepancy theorem with sharp applications to OEIS A284368–A284371

Research study prepared with ChatGPT, dated October 1, 2026; added to this report on 1 October 2026 in batch 7301 of ProveIt's incoming-report intake.

Sections 1–10 and Appendix A form the original report (Part I) of September 19, 2026, unchanged except for dated notes pointing to this addition, a front matter (title and abstract) that now announces both Parts, and the place of Appendix A, which now follows Part II together with Part II's own appendices. Part II was written as an extension of Part I's method to the family $0 \mapsto 1, 1 \mapsto 10^a 1^b$; it neither contains nor re-proves Part I's results. Section 11 records the provenance, fixes the notation of the addition, and compares the two families.

Summary (the manuscript's abstract). We study the fixed point of the binary substitution

$$\tau_{a,b}(0) = 1, \quad \tau_{a,b}(1) = 10^a 1^b, \quad a, b \geq 1.$$

Let

$$q = \frac{\sqrt{(b+1)^2 + 4a} - (b+1)}{2}.$$

The non-Perron eigenvalue of the incidence matrix is $-q$. We prove a sharp phase transition: prefix discrepancy, and hence the errors in the positions of zeros and ones, is bounded exactly when $a \leq b+1$. Throughout that bounded regime we determine the four exact state-dependent extrema. Writing

$$\widehat{C} = \frac{a-q}{1-q^2},$$

the closures of the position-error sets have endpoints

$$q - \widehat{C}, \quad q(1 + \widehat{C})$$

for positions of ones, and

$$\frac{1}{q} - \widehat{C} - 1, \quad \frac{C}{q}$$

for positions of zeros. None of these endpoints is attained, and all are subsequential limits.

The specializations $(a, b) = (1, 2)$ and $(2, 1)$ are OEIS A284368 and A284369. For A284368 we recover the sharper bounds

$$\frac{\sqrt{13}-5}{3} < E_1(n) < \frac{\sqrt{13}-2}{3}, \quad \frac{1+\sqrt{13}}{3} < E_0(n) < \frac{4+\sqrt{13}}{3}.$$

For A284369 we prove

$$-2 < E_1(n) < 1 + \sqrt{3}, \quad -\frac{3+\sqrt{3}}{2} < E_0(n) < 2 + \sqrt{3}.$$

The latter resolves the still-posted conjectures for its position sequences A284370 and A284371, with strictly stronger constants. In both OEIS cases the complete state attractors are intervals, so the displayed endpoints are not merely estimates but the exact error-value envelopes. We

also give constructive extremal subsequences, rational generating functions for their indices, a standard-library exact verifier, and a research program for the remaining interval-versus-Cantor and distribution questions.

Status and claim boundary. This Part extends the method of the existing ProveIt report on $0 \mapsto 1, 1 \mapsto (10)^m$ (Part I) rather than repackaging its A284364–A284366 results. The OEIS pages cited below still presented the A284368 and A284369 position statements as conjectures when checked on October 1, 2026. A284368 is already identified with the generalized s -Wythoff pair A184484–A184485, so its sharp floor formulas have an independent route and are included here as a calibration, not as the main novelty. The principal claims are the two-parameter theorem and the sharp A284369–A284371 envelopes. A limited source search found no proof of the A284369 bounds. No claim of exhaustive literature priority, peer review, or proof-assistant verification is made. The proofs here are self-contained, and the included computations use exact quadratic arithmetic.

11 Provenance, conventions and relation to Part I

Part II is built from one manuscript, number 44 of batch 73O1: *Exact position-error envelopes for quadratic morphic words* (October 1, 2026), delivered as OEIS_Quadratic_Morphic_Words_Research_Package.zip in arrival commit c79d64038 and placed in commit 9df4ba51a. It names no repository commit; it cites Part I by its repository URL. Its programs, data and figures are shipped with the prefix 02-morphic- (Appendix C and the README). The merge made these choices.

- Every theorem, corollary, proposition, lemma, remark, proof, table, figure and research question of the manuscript is printed. Its section n is Section $n + 11$ here, and its Appendices A, B are Appendices B, C.
- The manuscript cited Part I as “V. Reshetnikov, ProveIt repository, . . . , research draft”; Part I’s author line is “Research study prepared with ChatGPT”. The citation is replaced by references to Part I’s sections. The manuscript’s reference to Adamczewski’s 2004 paper is merged with Part I’s.
- The manuscript’s constant C is printed as $\widehat{C}$ (Table II.1); nothing else is renamed and no normalization changed.
- File names in the text are the shipped names; the delivered programs themselves still use their delivery names (Appendix C).

Notation. Part II uses Part I’s definitions of the prefix weight X , the next-letter states S_a , the counting function $O(N)$ and the position errors $E_0(n) = rn - u(n)$, $E_1(n) = sn - v(n)$, for its own word. Where a letter is shared with a different meaning, Table II.1 prints the Part I meaning beside the Part II one.

Remark 11.1 (The two families, compared). *The families are disjoint.* The word $(10)^m$ has the form $10^a 1^b$ only for $m = 1$, with $(a, b) = (1, 0)$, which Part II excludes ($a, b \geq 1$). So Part II neither contains, re-proves nor refines Part I’s theorems for A284364–A284366, and it does not say it does.

The same quadratic fields occur. For $(a, b) = (m, m - 1)$ the incidence matrix of $\tau_{a,b}$ is $\begin{pmatrix} 0 & m \\ 1 & m \end{pmatrix}$, the matrix of σ_m , so q, λ, r and s coincide; the words and the envelopes differ. For $m = 3$, Part I gives one-position errors in $(C - 3, C) \approx (-0.8835, 2.1165)$ and zero-position errors in

Table II.1: Symbols shared by the two Parts.

Symbol	Part I ($\sigma_m: 1 \mapsto (10)^m$)	Part II ($\tau_{a,b}: 1 \mapsto 10^a 1^b$)
q, λ, r, s	(2.2): $q = \frac{\sqrt{m^2+4m-m}}{2}, \lambda = m/q$	(13.3), (13.6): $q = \frac{\sqrt{(b+1)^2+4a-(b+1)}}{2}, \lambda = a/q$; r, s by the same formulas
$\widehat{C}$ vs C	$C = q/(1 - q^2)$ (2.3), the upper end of both position-error intervals	$\widehat{C} = (a - q)/(1 - q^2)$ (13.8), the manuscript's C ; it is the largest state-1 weight, the role played in Part I by H , not by C
H, t	(2.3); $t = 1 - q$	not used (t is a threshold in research question 3 of Section 21)
M	incidence matrix of σ_m	incidence matrix of $\tau_{a,b}$ (13.2)
I_0, I_1	the intervals (4.1)	the candidate hulls in each OEIS case, (17.2), (17.6)
K_0, K_1	any compact solution in Lemma 4.3	$K_i = \text{cl } S_i$ (Section 14)
P_j, P_k	extremal prefixes (7.1)	the different cycle (16.1)
T	$T = m(m + 2)$ (Corollary 7.2)	T_k , a prefix word (Section 18)
u	$u_m(n)$, position of the n th zero	$u(n)$ likewise; $u_i = \max K_i$ in Section 15 is a state extremum, not a position
F	$F_m(z)$, the characteristic series of the word	$F_0(n), F_1(n)$, the A284369 position errors

$((1 - C)/q, C) \approx (-1.4110, 2.1165)$, while Theorem 13.1 for $(a, b) = (3, 2)$ gives $(q - \widehat{C}, q(1 + \widehat{C})) \approx (-5.1165, 5.4661)$ and $(1/q - \widehat{C} - 1, \widehat{C}/q) \approx (-5.6440, 7.4661)$.

A formal consistency check. Setting $(a, b) = (1, 0)$ in the formulas of Theorem 13.1, outside its hypotheses, gives $q = (\sqrt{5} - 1)/2$, $\widehat{C} = q$, and the interval $(0, 1)$ for both letters: the complementary Beatty pair of Part I's case $m = 1$ (Section 9). Part II's proof does not cover this case (it uses $\widehat{C} - q^2(\widehat{C} + 1) = qb > 0$).

Method. Both Parts rest on the contracting weight $X = |P|_0 - q|P|_1$ with $X(\text{image}) = -qX$ and on states recording the next letter (Part I, Sections 3–4; Part II, Sections 14–15). Part II solves a four-variable Bellman system for the hulls and treats the question whether the attractors fill them separately (Section 17), whereas in Part I the intervals are invariant for every m .

Open questions. Part II's research question 3 (densities of ranks with error above a threshold) is the question Part I leaves open in Section 9; it is not answered in either Part.

12 The OEIS questions and the new result

12.1 Two neighboring morphic words

The fixed point of

$$0 \mapsto 1, \quad 1 \mapsto 1011 \quad (12.1)$$

is A284368. Its entry identifies the positions of ones and zeros with A184484 and A184485, respectively, and conjectures [6, 7, 8]

$$1 < n \frac{5 + \sqrt{13}}{2} - u(n) < 3, \quad -1 < n \frac{\sqrt{13} - 1}{2} - v(n) < 1. \quad (12.2)$$

Here and below $u(n)$ is the position of the n th zero and $v(n)$ the position of the n th one. The older entries A184484–A184485 describe the same complementary pair as an s -Wythoff

pair. Thus some consequences can also be approached through generalized Beatty theory; our purpose is an exact substitution-state analysis that extends uniformly to a larger family.

The neighboring fixed point

$$0 \mapsto 1, \quad 1 \mapsto 1001 \quad (12.3)$$

is A284369 [9]. The parent entry and the zero-position entry A284370 conjecture [10]

$$-3 < n \frac{3 + \sqrt{3}}{2} - u(n) < 4, \quad (12.4)$$

while the parent entry gives the one-position conjecture

$$-2 < n\sqrt{3} - v(n) < 3. \quad (12.5)$$

The standalone A284371 page currently omits the term $-a(n)$ in its printed inequality, but its title, complement relation, and the parent entry make the intended statement (12.5) unambiguous [11]. [Added 1 October 2026, batch 73O1: this describes the A284371 page as the manuscript saw it on 1 October 2026; it was not re-checked at intake.]

12.2 Specific resolution

Theorem 12.1 (Sharp OEIS envelopes). *For the word A284368, put*

$$E_0(n) = n \frac{5 + \sqrt{13}}{2} - u(n), \quad E_1(n) = n \frac{\sqrt{13} - 1}{2} - v(n).$$

Then, for every $n \geq 1$,

$$\frac{1 + \sqrt{13}}{3} < E_0(n) < \frac{4 + \sqrt{13}}{3}, \quad (12.6)$$

$$\frac{\sqrt{13} - 5}{3} < E_1(n) < \frac{\sqrt{13} - 2}{3}. \quad (12.7)$$

The closures of the two error-value sets are the corresponding closed intervals, and every displayed endpoint is a subsequential limit. Consequently both conjectures in (12.2) hold, with substantial room to spare.

For the word A284369, put

$$F_0(n) = n \frac{3 + \sqrt{3}}{2} - u(n), \quad F_1(n) = n\sqrt{3} - v(n).$$

Then, for every $n \geq 1$,

$$-\frac{3 + \sqrt{3}}{2} < F_0(n) < 2 + \sqrt{3}, \quad (12.8)$$

$$-2 < F_1(n) < 1 + \sqrt{3}. \quad (12.9)$$

Again the closures are exactly the corresponding closed intervals, with all four endpoints occurring as subsequential limits. Hence (12.4)–(12.5) are true, and the constants in them can be sharpened to those above.

The A284368 intervals have width exactly one, so they also recover explicit floor formulas.

Corollary 12.2 (The generalized-Wythoff specialization). *For A284368,*

$$v(n) = \left\lfloor n \frac{\sqrt{13} - 1}{2} + \frac{5 - \sqrt{13}}{3} \right\rfloor, \quad (12.10)$$

$$u(n) = \left\lfloor n \frac{5 + \sqrt{13}}{2} - \frac{1 + \sqrt{13}}{3} \right\rfloor. \quad (12.11)$$

These formulas agree with the OEIS identification of the two position sequences as A184484–A184485; no independent priority claim is made for this corollary.

Proof. For example, (12.7) says that $0 < n(\sqrt{13} - 1)/2 + (5 - \sqrt{13})/3 - v(n) < 1$, which is equivalent to (12.10). The zero formula follows identically from (12.6). $\square$

The decimal endpoints are summarized in Table II.2.

Table II.2: Conjectured and sharp position-error envelopes.

word	letter	OEIS interval	exact sharp interval
A284368	0	(1, 3)	(1.5351837584..., 2.5351837584...)
A284368	1	(−1, 1)	(−0.4648162415..., 0.5351837584...)
A284369	0	(−3, 4)	(−2.3660254037..., 3.7320508075...)
A284369	1	(−2, 3)	(−2, 2.7320508075...)

The proof comes from a two-parameter theorem. It is close in spirit to the state-dependent contraction used in Part I for $1 \mapsto (10)^m$ (Sections 3–4), but the internal block geometry is different and produces a distinct Bellman system.

13 The two-parameter family

Fix integers $a, b \geq 1$ and define

$$\tau_{a,b}(0) = 1, \quad \tau_{a,b}(1) = 10^a 1^b. \quad (13.1)$$

Because the image of 1 begins with 1, the iterates $\tau_{a,b}^k(1)$ are nested prefixes of a fixed infinite word $\mathbf{w}^{(a,b)}$. The substitution is primitive. The relation between subdominant substitution eigenvalues and discrepancy is classical [5, 4]; the point here is to solve the legal-prefix state system exactly for this family.

Its incidence matrix, with rows and columns ordered 0, 1, is

$$M = \begin{pmatrix} 0 & a \\ 1 & b + 1 \end{pmatrix}. \quad (13.2)$$

Set

$$q = \frac{\sqrt{(b+1)^2 + 4a} - (b+1)}{2}, \quad \lambda = b + 1 + q = \frac{a}{q}. \quad (13.3)$$

Then

$$q^2 + (b+1)q = a, \quad (13.4)$$

and the eigenvalues of M are λ and $-q$. A positive right Perron eigenvector is $(q, 1)^T$, so the zero and one frequencies are

$$\delta = \frac{q}{1+q}, \quad \rho = \frac{1}{1+q}. \quad (13.5)$$

The corresponding position slopes are

$$r = \frac{1}{\delta} = 1 + \frac{1}{q}, \quad s = \frac{1}{\rho} = 1 + q. \quad (13.6)$$

The simple inequality

$$q < 1 \iff \sqrt{(b+1)^2 + 4a} < b+3 \iff a < b+2$$

gives, for integer parameters,

$$q < 1 \iff a \leq b+1. \quad (13.7)$$

This is exactly the bounded-discrepancy regime.

Theorem 13.1 (General sharp envelope theorem). *Assume $a, b \geq 1$ and $a \leq b+1$. Define*

$$\widehat{C} = \frac{a-q}{1-q^2}. \quad (13.8)$$

Let $u(n)$ and $v(n)$ be the positions of the n th zero and one in $\mathbf{w}^{(a,b)}$, and let

$$E_0(n) = \left(1 + \frac{1}{q}\right)n - u(n), \quad E_1(n) = (1+q)n - v(n).$$

Then for every $n \geq 1$,

$$\frac{1}{q} - \widehat{C} - 1 < E_0(n) < \frac{C}{q}, \quad (13.9)$$

$$q - \widehat{C} < E_1(n) < q(1 + \widehat{C}). \quad (13.10)$$

Each endpoint is sharp and is approached by an explicit infinite subsequence. No endpoint is attained at a finite index.

The centered one-count discrepancy satisfies, for every prefix length $N \geq 0$,

$$-\frac{\widehat{C}}{1+q} < O(N) - \frac{N}{1+q} < \frac{q(\widehat{C}+1)}{1+q}, \quad (13.11)$$

where $O(N)$ counts ones in the first N letters. Both constants are sharp. In particular the densities in (13.5) exist.

Theorem 13.2 (Spectral phase transition). *For the family (13.1), the prefix discrepancy is bounded if and only if $a \leq b+1$. If $a = b+2$, there are prefixes whose discrepancy grows linearly with substitution depth, hence at least logarithmically with prefix length. If $a \geq b+3$, there are prefixes whose discrepancy grows geometrically with substitution depth.*

Sections 14–15 prove these theorems. The specific interval attractors needed for Theorem 12.1 are resolved in Section 17.

14 Contracting weights and exact prefix states

For a finite binary word P , define the quadratic weight

$$X(P) = |P|_0 - q|P|_1. \quad (14.1)$$

It is additive under concatenation.

Lemma 14.1 (Weight eigenrelation). *For every finite word P ,*

$$X(\tau_{a,b}(P)) = -qX(P). \quad (14.2)$$

Proof. For the letter 0, the two sides both equal $-q$. For the letter 1,

$$X(10^a1^b) = a - q(b+1) = q^2 = -qX(1),$$

using (13.4). Additivity proves the identity for any word. $\square$

A prefix must remember the letter immediately following it. Let

$$S_i = \{X(P) : P \text{ is a prefix of } \mathbf{w}^{(a,b)} \text{ and the next letter is } i\}, \quad i \in \{0, 1\}, \quad (14.3)$$

and let $K_i = \text{cl } S_i$.

Lemma 14.2 (Graph-directed state equations). *If $q < 1$, the compact sets K_0, K_1 are the unique nonempty compact solution of*

$$K_0 = \bigcup_{j=0}^{a-1} (-qK_1 + j - q), \quad (14.4)$$

$$K_1 = (-qK_0) \cup (-qK_1) \cup \bigcup_{\ell=1}^b (-qK_1 + a - \ell q). \quad (14.5)$$

Proof. Let P be a parent prefix and suppose the next parent letter is 1. Inside its image 10^a1^b , the prefixes immediately before the zeros are

$$1, 10, \dots, 10^{a-1},$$

with weights $j - q$ for $0 \leq j < a$. The prefixes immediately before the initial one and the b trailing ones have weights

$$0, \quad a - q, \quad a - 2q, \quad \dots, \quad a - bq.$$

If the next parent letter is 0, its image is the single letter 1, so the only transition is to state 1 with internal weight 0. Combining these internal weights with Lemma 14.1 gives (14.4)–(14.5).

Conversely, every position lies in one of these image blocks, so no transitions are missing. Since all affine maps have contraction ratio $q < 1$, the graph-directed Hutchinson operator is a contraction in the maximum Hausdorff metric. It therefore has a unique compact fixed pair, and the closures of the actual prefix-state sets form that pair. $\square$

The relation

$$a - bq = q(q+1) > 0 \quad (14.6)$$

follows from (13.4). Thus the smallest state-1 internal digit is 0 and the largest is $a - q$. This observation makes the extrema solvable in closed form even when the attractors themselves have gaps.

15 Solving the Bellman extremum system

Write

$$\ell_i = \min K_i, \quad u_i = \max K_i.$$

Because multiplication by $-q$ reverses order, equations (14.4)–(14.5) give

$$\ell_0 = -qu_1 - q, \quad u_0 = -q\ell_1 + a - 1 - q, \quad (15.1)$$

$$\ell_1 = \min\{-qu_0, -qu_1\}, \quad u_1 = \max\{-q\ell_0, -q\ell_1 + a - q\}. \quad (15.2)$$

Proposition 15.1 (Exact state extrema). *In the bounded regime $q < 1$, with $\widehat{C}$ as in (13.8),*

$$\boxed{\ell_0 = -q(\widehat{C} + 1), \quad u_0 = \widehat{C} - 1, \quad \ell_1 = -qC, \quad u_1 = \widehat{C}.} \quad (15.3)$$

None of these four values belongs to the corresponding unclosed state set S_i , but each is a limit of values in that state.

Proof. The extremum map defined by the right sides of (15.1)–(15.2) is q -Lipschitz in the supremum norm, because minima and maxima are nonexpansive. It therefore has at most one fixed point.

We verify the displayed candidate. Since $u_1 = \widehat{C} > \widehat{C} - 1 = u_0$,

$$\ell_1 = -qu_1 = -qC.$$

Also

$$-q\ell_1 + a - q = q^2\widehat{C} + a - q = \widehat{C}$$

by $\widehat{C}(1 - q^2) = a - q$. The competing maximum branch is

$$-q\ell_0 = q^2(\widehat{C} + 1) < \widehat{C},$$

because

$$\widehat{C} - q^2(\widehat{C} + 1) = a - q - q^2 = qb > 0.$$

Thus $u_1 = \widehat{C}$. The formulas for ℓ_0 and u_0 now follow directly; for the latter,

$$-q\ell_1 + a - 1 - q = q^2\widehat{C} + a - q - 1 = \widehat{C} - 1.$$

Uniqueness proves (15.3).

For nonattainment, suppose first that an actual state-1 prefix had weight $\widehat{C}$. The strict inequality between the two maximum branches forces its parent to have state-1 weight $-qC$ and to use the digit $a - q$. Attaining $-qC$ in turn forces a parent of state 1 and weight $\widehat{C}$, using digit 0. Two desubstitutions therefore return to the same extremal requirement while strictly shortening a nonempty prefix, an infinite-descent contradiction. The same argument starts from $-qC$. Attainment of either state-0 endpoint would force attainment of the corresponding state-1 endpoint through (14.4). Finally, iterating the two extremizing transitions from the empty prefix constructs sequences converging to the endpoints, so they are sharp. $\square$

15.1 Counting discrepancy

If P has length N , then

$$X(P) = N - (1 + q)O(N), \quad O(N) - \frac{N}{1 + q} = -\frac{X(P)}{1 + q}. \quad (15.4)$$

Across both states, Proposition 15.1 gives

$$-q(\widehat{C} + 1) < X(P) < \widehat{C},$$

which is exactly (13.11).

15.2 Position errors

Suppose a prefix P is immediately followed by the n th one. Then $|P|_1 = n - 1$ and $|P| = v(n) - 1$, so

$$E_1(n) = (1 + q)n - v(n) = q - X(P). \quad (15.5)$$

Because $X(P) \in S_1$, Proposition 15.1 yields

$$q - \widehat{C} < E_1(n) < q + qC = q(1 + \widehat{C}).$$

Similarly, immediately before the n th zero one has $|P|_0 = n - 1$ and $|P| = u(n) - 1$, whence

$$E_0(n) = \left(1 + \frac{1}{q}\right)n - u(n) = \frac{X(P) + 1}{q}. \quad (15.6)$$

Using $X(P) \in S_0$ gives

$$\frac{1}{q} - \widehat{C} - 1 < E_0(n) < \frac{C}{q}.$$

This proves Theorem 13.1.

16 The bounded-discrepancy phase transition

The preceding contraction proof establishes bounded discrepancy for $q < 1$. To prove that this condition is necessary, one explicit state-1 cycle is enough.

Let P_0 be the empty prefix and recursively define

$$P_{k+1} = \tau_{a,b}^2(P_k) 1 0^a. \quad (16.1)$$

Every P_k is a prefix immediately before a one. If $x_k = X(P_k)$, then

$$x_0 = 0, \quad x_{k+1} = q^2 x_k + a - q. \quad (16.2)$$

The two steps are: use the zero internal digit before the initial one, then use the largest state-1 digit $a - q$ before the first trailing one.

When $q < 1$,

$$x_k = \widehat{C}(1 - q^{2k}) \longrightarrow \widehat{C}, \quad (16.3)$$

which also gives an explicit sharpness sequence. When $q = 1$,

$$x_k = k(a - 1) = k(b + 1),$$

so discrepancy is unbounded linearly in depth. When $q > 1$, the solution of (16.2) grows geometrically like q^{2k} . Together with (13.7), this proves Theorem 13.2.

At the boundary $q = 1$, substitution lengths grow exponentially in k while x_k grows linearly, giving a logarithmic lower bound in the physical prefix length. A matching $O(\log N)$ upper bound follows by desubstituting any prefix: every twofold image has length at least 3, each internal correction is bounded, and the multiplier has absolute value one. For $q > 1$, the constructed prefixes satisfy $|P_k| \asymp \lambda^{2k}$ and $X(P_k) \asymp q^{2k}$, hence exhibit the power law $X(P_k) \asymp |P_k|^{\log_\lambda q}$. A full normalized limit theory is left for future work. These refinements are not needed for the OEIS applications.

17 When the state attractors are full intervals

Proposition 15.1 determines the hulls of K_0, K_1 for every bounded parameter pair. It does not assert that the compact sets fill their hulls. Indeed, for some larger parameters the first-level images already leave gaps. The two OEIS cases are especially tractable: one is an exact tiling and the other has ample overlap.

17.1 A284368: an exact interval tiling

Here $(a, b) = (1, 2)$,

$$q = \frac{\sqrt{13}-3}{2}, \quad \widehat{C} = \frac{\sqrt{13}+1}{6}. \quad (17.1)$$

Proposition 15.1 gives the candidate intervals

$$I_0 = \left[\frac{2-\sqrt{13}}{3}, \frac{\sqrt{13}-5}{6} \right], \quad (17.2)$$

$$I_1 = \left[\frac{\sqrt{13}-5}{6}, \frac{\sqrt{13}+1}{6} \right]. \quad (17.3)$$

The state-0 equation has only one image, $I_0 = -qI_1 - q$. The four state-1 images tile I_1 exactly, in the order shown in Table II.3.

Table II.3: Exact interval tiling for $(a, b) = (1, 2)$. Consecutive endpoints coincide.

source and digit	left endpoint	right endpoint
$-qI_1$	$(\sqrt{13}-5)/6$	$-7/3 + 2\sqrt{13}/3$
$-qI_0$	$-7/3 + 2\sqrt{13}/3$	$19/6 - 5\sqrt{13}/6$
$-qI_1 + 1 - 2q$	$19/6 - 5\sqrt{13}/6$	$5/3 - \sqrt{13}/3$
$-qI_1 + 1 - q$	$5/3 - \sqrt{13}/3$	$(\sqrt{13}+1)/6$

Thus (I_0, I_1) satisfies the graph-directed equations as sets. Uniqueness in Lemma 14.2 proves

$$K_0 = I_0, \quad K_1 = I_1. \quad (17.4)$$

Applying the affine error maps (15.5)–(15.6) gives exactly (12.6)–(12.7) and proves the A284368 half of Theorem 12.1.

17.2 A284369: overlapping interval covers

Here $(a, b) = (2, 1)$,

$$q = \sqrt{3} - 1, \quad \widehat{C} = 1 + \sqrt{3}. \quad (17.5)$$

The state hulls are

$$I_0 = [-1 - \sqrt{3}, \sqrt{3}], \quad I_1 = [-2, 1 + \sqrt{3}]. \quad (17.6)$$

For state 0, the two image intervals are

$$[-1 - \sqrt{3}, \sqrt{3} - 1], \quad [-\sqrt{3}, \sqrt{3}], \quad (17.7)$$

which overlap and cover I_0 . For state 1, the three images are

$$[-2, 2\sqrt{3} - 2], \quad [\sqrt{3} - 3, 2], \quad [1 - \sqrt{3}, 1 + \sqrt{3}]. \quad (17.8)$$

They overlap consecutively and cover I_1 . Therefore the candidate pair is again an exact set-valued fixed point, and uniqueness gives

$$K_0 = I_0, \quad K_1 = I_1. \quad (17.9)$$

The position-error maps yield (12.8)–(12.9). This completes the proof of Theorem 12.1.

Remark 17.1. The interval property is not automatic. For example, at $(a, b) = (2, 5)$ the two first-level state-0 images have a gap, although the extremal formulas of Proposition 15.1 remain valid. Classifying exactly when all gaps close is one of the main open directions in Section 21.

18 Constructive extremal subsequences and generating functions

The cycle (16.1) not only proves sharpness but gives explicit indices. Let

$$c_k = \begin{pmatrix} z_k \\ o_k \end{pmatrix} = \begin{pmatrix} |P_k|_0 \\ |P_k|_1 \end{pmatrix}. \quad (18.1)$$

Since the suffix 10^a has count vector $(a, 1)^T$,

$$c_0 = 0, \quad c_{k+1} = M^2 c_k + \begin{pmatrix} a \\ 1 \end{pmatrix}. \quad (18.2)$$

The next one has rank $o_k + 1$ and position $z_k + o_k + 1$. Its error is $q - x_k$, which decreases to $q - \widehat{C}$. To exhibit all four endpoints explicitly, put

$$Q_k = \tau_{a,b}(P_k), \quad R_k = \tau_{a,b}(P_k)1, \quad T_k = \tau_{a,b}(Q_k)10^{a-1}.$$

Then Q_k is in state 1 with weight $-qx_k \rightarrow -qC$, so its one-position errors tend to $q(1 + \widehat{C})$. The prefixes R_k and T_k are in state 0 and have respective weights

$$-qx_k - q \rightarrow -q(\widehat{C} + 1), \quad -qX(Q_k) + a - 1 - q \rightarrow \widehat{C} - 1.$$

Their zero-position errors therefore tend to $1/q - \widehat{C} - 1$ and $\widehat{C}/q$. This supplies concrete infinite subsequences for every endpoint in Theorem 13.1.

The vector generating function is rational:

$$\sum_{k \geq 0} c_k z^k = \frac{z}{1 - z} (I - zM^2)^{-1} \begin{pmatrix} a \\ 1 \end{pmatrix}. \quad (18.3)$$

Let $N_k = z_k + o_k = |P_k|$. In the two OEIS cases,

$$\sum_{k \geq 0} N_k z^k = \frac{z(2 - 5z)}{(1 - z)(1 - 11z + z^2)} \quad \text{for A284368,} \quad (18.4)$$

$$\sum_{k \geq 0} N_k z^k = \frac{3z(1 - 2z)}{(1 - z)(1 - 8z + 4z^2)} \quad \text{for A284369.} \quad (18.5)$$

The first terms are

A284368: 0, 2, 19, 204, 2222, 24235, 264360, 2883722, ... ,

A284369: 0, 3, 21, 153, 1137, 8481, 63297, 472449,

Thus the corresponding homogeneous recurrences are

$$N_k = 12N_{k-1} - 12N_{k-2} + N_{k-3} \quad \text{for A284368,}$$

$$N_k = 9N_{k-1} - 12N_{k-2} + 4N_{k-3} \quad \text{for A284369.}$$

These formulas make the sharpness sequences reproducible without building the full fixed words.

19 Computational audit

The verifier code/`02-morphic-verify.py` uses only the Python standard library. It represents every algebraic quantity as $A + B\sqrt{D}$ with rational coefficients, compares signs by integer arithmetic, and never uses a decimal to decide an inequality. Its main checks are:

1. the general Bellman solution for all 119 parameter pairs $1 \leq b \leq 14, 1 \leq a \leq b + 1$;
2. exact interval-tiling and overlap identities for the two OEIS cases;
3. literal generation and comparison with the posted initial OEIS terms;
4. exact checking of every zero- and one-position error in the first 100 000 letters of each word;
5. the extremal vector recurrences and data tables used above.

The recorded run (`data/02-morphic-verification.json`) reports 201 686 exact or counting checks. This finite audit is not used in place of the proofs; it is intended to catch indexing, letter-order, and algebraic-simplification errors.

Figures 2 and 3 plot the first 180 000 letters. The dashed lines are the exact endpoints, not fitted values. The apparent filling of the vertical bands is explained by the interval-attractor proofs in Section 17.

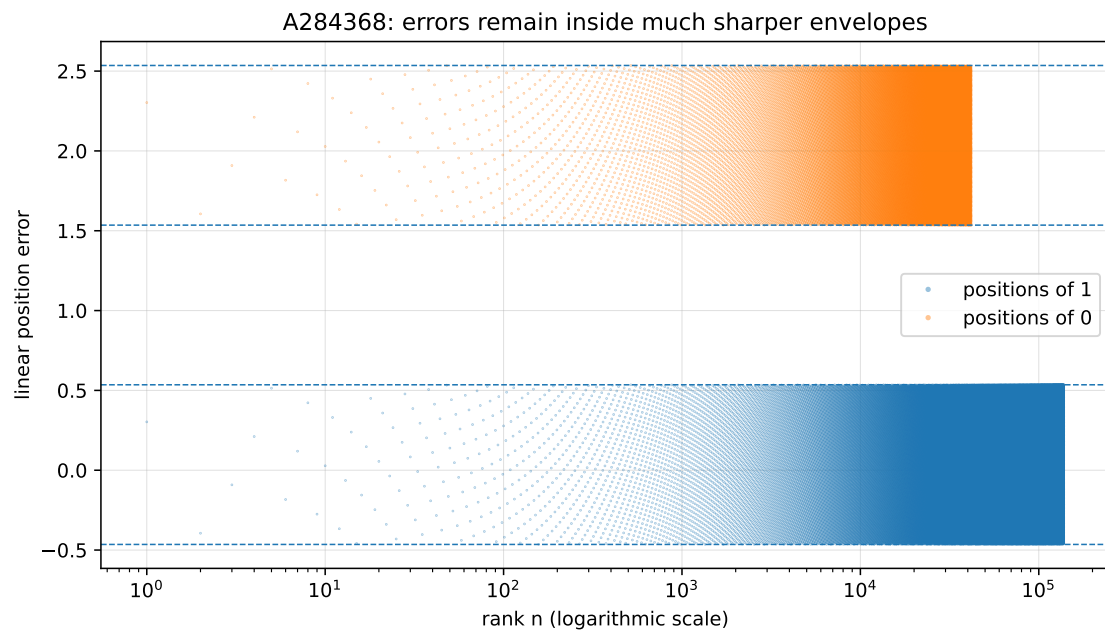


Figure 2: Position errors for A284368. The zero and one errors lie in two unit-width sharp intervals; the displayed envelope constants are exact.

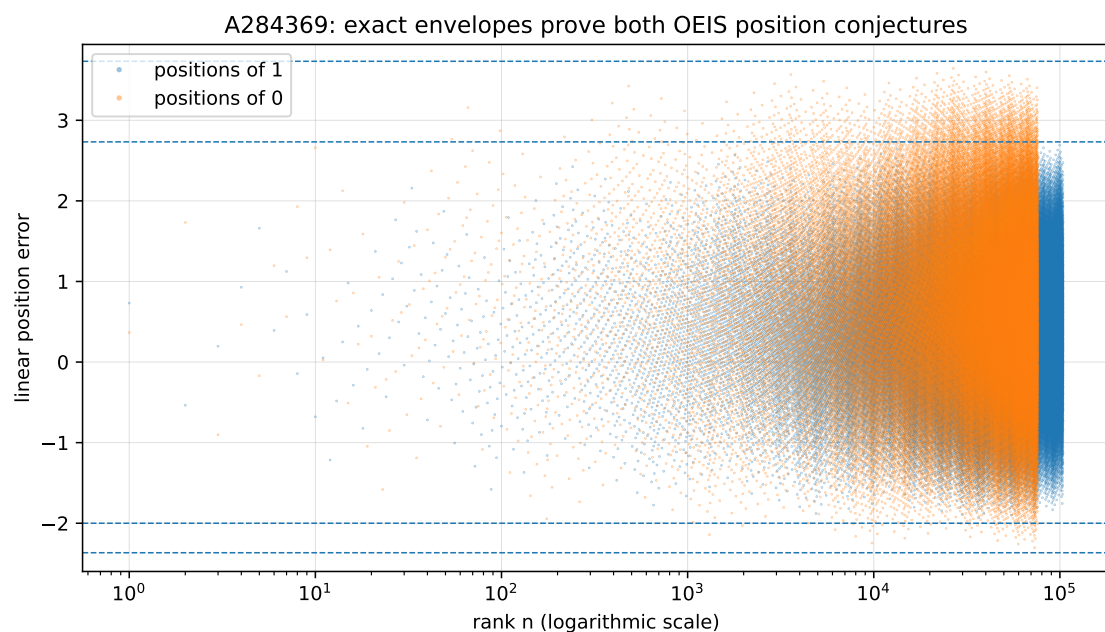


Figure 3: Position errors for A284369. The lower one-error endpoint is exactly -2 but is never attained, proving the strict OEIS inequality.

20 OEIS-ready consequences

The following concise statements can be adapted into OEIS comments after independent review.

A284368 / A184484 / A184485. For the fixed point of $0 \mapsto 1, 1 \mapsto 1011$, let $u(n)$ and $v(n)$ be the positions of the n th zero and one. Then

$$\frac{1 + \sqrt{13}}{3} < n \frac{5 + \sqrt{13}}{2} - u(n) < \frac{4 + \sqrt{13}}{3},$$

$$\frac{\sqrt{13} - 5}{3} < n \frac{\sqrt{13} - 1}{2} - v(n) < \frac{\sqrt{13} - 2}{3}.$$

All four constants are sharp but unattained. The zero and one densities are $(5 - \sqrt{13})/6$ and $(1 + \sqrt{13})/6$.

A284369 / A284370 / A284371. For the fixed point of $0 \mapsto 1, 1 \mapsto 1001$, let $u(n)$ and $v(n)$ be the positions of the n th zero and one. Then

$$-\frac{3 + \sqrt{3}}{2} < n \frac{3 + \sqrt{3}}{2} - u(n) < 2 + \sqrt{3},$$

$$-2 < n\sqrt{3} - v(n) < 1 + \sqrt{3}.$$

All four constants are sharp but unattained. The zero and one densities are $1 - 1/\sqrt{3}$ and $1/\sqrt{3}$.

No OEIS edit has been submitted as part of this work.

21 Further research questions

The two-state reduction exposes several problems that are both concrete and substantially broader than the four resolved inequalities.

1. **Interval-versus-Cantor classification.** Determine necessary and sufficient conditions on (a, b) , with $a \leq b + 1$, for K_0 and K_1 to equal their convex hulls. First-level overlap gives explicit inequalities, but deeper exact overlaps can close apparent gaps. A finite-state classification may exist because all translations lie in the same quadratic field.
2. **Invariant error distributions.** The present paper identifies the support and its endpoints. Determine the limiting empirical measures of $E_0(n)$ and $E_1(n)$, their continuity or singularity, moments, and local dimensions. In interval cases, decide whether the measures are absolutely continuous or have self-similar density singularities.
3. **Threshold densities and record statistics.** For a fixed threshold t , determine the natural density of ranks with $E_i(n) > t$, when it exists. This directly continues an open direction in Part I (A284364–A284366; Section 9), where sharp supports are known but violation densities are not.
4. **Exact first-hitting algorithms.** Develop automata or Ostrowski-type numeration systems that return the first rank at which a prescribed algebraic error interval is entered. The graph system suggests a branch-and-bound algorithm with exact quadratic endpoints.

5. **Closed forms for A284370 and A284371.** A284368 is linked to generalized Wythoff sequences. Determine whether the A284369 position sequences admit finite generalized-Beatty formulas, nested floor formulas, or low-complexity transducers. The broad interval support makes a single ordinary Beatty formula impossible, but structured finite combinations may remain possible.
6. **Factor balance rather than prefix balance.** Compute the optimal constant controlling the difference in letter counts between arbitrary equal-length factors. Prefix envelopes give immediate coarse bounds, but the sharp factor constant requires the difference set $K_i - K_j$ together with legal endpoint states.
7. **Critical and supercritical normalization.** At $a = b + 2$, determine the exact limsup and liminf after division by $\log N$. For $a \geq b + 3$, normalize by the predicted power $N^{\log_\lambda q}$ and identify the resulting graph-directed limit sets.
8. **Automated OEIS survey.** Apply the state-equation pipeline to other primitive binary morphisms whose OEIS entries contain unproved Beatty-type error bounds. The steps—incidence spectrum, eigenweight, legal internal digits, Bellman extrema, and overlap audit—are sufficiently explicit to automate candidate discovery while retaining exact proof certificates.
9. **Formalization in Lean.** Formalize finite-word counts, substitution desubstitution, quadratic-field order, and the contracting extremum map in ProveIt. The compact-attractor statement can either be formalized directly or replaced, for the extremal bounds, by finite-depth inequalities plus a geometric tail estimate.

22 Conclusion

The position conjectures for A284369 are not isolated numerical accidents. They are the $(a, b) = (2, 1)$ specialization of a two-parameter theorem whose constants are forced by a four-variable contracting Bellman system. The same theorem gives an independent, sharper resolution of the neighboring A284368 bounds and reveals a clean spectral transition at $a = b + 2$.

The main structural lesson is that the non-Perron eigenvalue gives the contraction rate, but sharp constants require legal prefix states. Once those states are retained, the exact extrema collapse to the single algebraic constant $\widehat{C} = (a - q)/(1 - q^2)$. Whether the resulting compact sets fill their hulls is a separate overlap problem: it succeeds by exact tiling for A284368, by overlap for A284369, and fails already for some nearby parameters. That separation between sharp envelopes and internal fractal geometry provides a natural next research program.

Appendix A belongs to Part I (and is printed as it was); Appendices B and C belong to Part II.

A Reproducing and inspecting the deliverables

All verification code uses only the Python standard library and requires Python 3.10 or later. From the archive’s top-level directory run:

```
python code/independent_check.py
python code/verify.py --long
```

The shorter command `python code/verify.py` omits the 20-million-letter literal construction but retains the exact first-counterexample search and finite certificate. A successful run writes its own report and regenerates the exact data. Running the shorter command will replace the saved report with one that accurately records that the long test was omitted.

For an individual large-index query:

```
from substitution import Substitution

model = Substitution(3)
assert model.select(1, 2977771) == 5334043
assert model.select(0, 8933313) == 20222898
print(model.first_at_least(1, threshold=2))
```

Run this from `code/`, or add that directory to the Python module search path. Ranks and returned positions are 1-based; prefix lengths and internal offsets are 0-based. The exact search’s `None` result means that no witness was found through the specified maximum block level, not that the infinite sequence has no witness.

The main archive components are described in Table 3. The pre-rendered figure is included, so compiling the article does not require Python or Matplotlib. To rebuild the PDF, run `pdflatex` on `article.tex` twice; the bibliography is embedded in the source. To rebuild only the plot, run `python code/make_figure.py`, which requires Matplotlib. No fonts, checksum files, temporary TeX build files, or downloaded copies of other authors’ papers are included.

File or directory	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Complete source and compiled article.
<code>code/substitution.py</code>	Exact quadratic arithmetic, summaries, counts, selection, first-threshold search.
<code>code/independent_check.py</code>	Separate compact checker for the finite certificate.
<code>code/verify.py</code>	Executable cross-checks and data generation.
<code>data/certificates.json</code>	First-counterexample traces and exact certificate coefficients.
<code>data/finite_extrema.json</code>	Block extrema through W_{30} .
<code>data/extremal_family.json</code>	First 25 explicit extremal-family terms.
<code>data/verification.json</code>	Report of the executed test suite.
<code>figures/</code>	The figure in vector PDF and raster PNG form.
<code>oeis_proposed_updates.txt</code>	Suggested mathematical corrections, not submitted to OEIS.
<code>README.md</code>	Build instructions, scope, and verification notes.

Table 3: Archive contents.

B A compact proof checklist

For ease of independent checking, the argument can be reduced to the following algebraic chain.

1. $q^2 + (b + 1)q = a$ and $X = \#0 - q\#1$ imply $X(\tau(P)) = -qX(P)$.
2. Legal internal prefixes give exactly (14.4)–(14.5).
3. Since $a - bq = q(q + 1) > 0$, the extreme state-1 digits are 0 and $a - q$.
4. The extremum operator is q -contractive, and

$$(-q(\widehat{C} + 1), \widehat{C} - 1, -qC, \widehat{C})$$

is its fixed point because $\widehat{C}(1 - q^2) = a - q$ and $\widehat{C} - q^2(\widehat{C} + 1) = qb > 0$.

5. Immediately before the n th one and zero, the error maps are $q - X$ and $(X + 1)/q$, respectively.
6. The two-step extremal cycle has affine map $x \mapsto q^2x + a - q$, proving sharpness and the phase transition.

C Reproducibility files

The archive delivered with the manuscript contained the files below. The manuscript's `article.tex`, `article.pdf`, `README.md` and the file list `MANIFEST.txt` are not shipped (the text is Part II of this report; its checksum ledger was verified and retired). The other files are shipped byte-identical under the prefix `02-morphic-`:

shipped file	purpose
<code>code/02-morphic-verify.py</code>	exact standard-library verifier
<code>code/02-morphic-make_figures.py</code>	deterministic figure generator
<code>code/02-morphic-build.sh</code>	the manuscript's PDF build script
<code>data/02-morphic-verification.json</code>	recorded exact-check output
<code>figures/02-morphic-A28436*_errors.{pdf,png}</code>	vector and raster figures
<code>02-morphic-oeis_proposed_updates.txt</code>	concise draft comments, not submitted
<code>02-morphic-research_status.md</code>	novelty and verification boundaries

The programs still use their delivery names: the verifier writes `data/verification.json`, which in this report is *Part I*'s recorded report; the figure script writes unprefixes `figures/A28436*_errors.*`; and the build script builds `article.tex`, which here is the whole report. Run them on a copy of the delivered layout (see the README).

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